

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-a0e9bd1e0b830bf21.jsonl`  
- SHA-256: `736fc50a37a4746a91782c65fa5749cb9a5dd0961ec87618ea9c6955128c6674`  
- Source modified: `2026-06-19T17:19:27+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_4e32c8b8-725`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T17:18:10.922Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:

- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): CODE_MAP.md, EVALUATION.md, HOW_RALLY_DETECTION_WORKS.md, NEXT_STEPS.md, ROADMAP.md, BACKLOG.md, PROJECT_DOC_TEMPLATE.md

For EACH doc:

1. Read the status header + TL;DR/intro + (if present) the \$7/\$8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.

5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:15.568Z)

I'll audit the docs in my batch. Let me start by reading the archive policy and the docs themselves, plus checking for already-archived copies.

3. user (2026-06-19T17:18:16.521Z)

```
1      # archives/
2
3      Umbrella directory for all **historical and completed** material.
4
5      ## Live vs Historical
6      - **Live docs** (for ramp-up, constant updates, and current work) live at the top level
of `docs/`.
7      - Everything here is no longer the primary source of truth. It is kept for context,
decision history, and regression detection.
8
9      ## Classification inside archives/
10     - `past_projects/` ? Completed small tech debt, starter projects, or low-regret
initiatives (e.g. the original low-regret moves + scaffolding work). These are "past" once they
reach a terminal state.
11     - `decisions/` ? Historical ADRs and major architectural decisions.
12     - `research/` ? Completed or superseded research notes and plans.
13     - `checkpoints/` ? Snapshots of major project states (especially useful for detecting
regressions in quality, priorities, or baselines over time as we do quality iterations).
14     - Other one-off audits, etc.
15
16     ## Archival Rule
17     Only move something into `archives/` (or its subdirectories) when the work is in a clear
**terminal state**:
18     - DONE
19     - COMPLETED
20     - REJECTED
21
22     Never archive in-flight or pending work.
23
24     When archiving a project/phase:
25     1. Ensure the main document has an explicit **STATUS: COMPLETED** (or DONE/REJECTED)
header at the top.
26     2. Link back to the current living documents in `../` (e.g. `COMMERCIALIZATION.md`,
`BACKLOG.md`).
27     3. For regression detection value, consider adding a dated entry under `checkpoints/`.
28
29     See `past_projects/README.md` for the specific guidance on that category.
30
```

4. user (2026-06-19T17:18:16.903Z)

```
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
```

docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\decisions\DECISIONS.md
docs\archives\TEST_RUN_SUMMARY.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
docs\archives\quality-iterations\README.md

5. assistant (2026-06-19T17:18:19.331Z)

Already-archived copies exist for three hardening docs (none in my batch). Now let me read the headers/intros of my batch docs.

6. user (2026-06-19T17:18:20.559Z)

```
1      # CODE_MAP.md
2
3      **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-
hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a
segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.
4
5      **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this
doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure
(this revision follows #40).
6
7      ? **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust
git history over stale lines until the next full regen:** `add_video` is now an **UPSERT** (no
FK-cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via
`segment_watermarks`/`prune_segments`); detector artifacts are keyed by **video_id**, not
basename; **`indexing.wasb.*` is now a real typed config** (was silently dropped); `timeout_sec`
defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**;  
`match_intervals` requires real overlap (iou>0); `run_regression` baseline lives in tracked
`eval_baselines/` and **exits 3 on a missing baseline**;  
the API validates `video_path`/`output_filename`. New tests: `test_api_endpoints`, `test_compiler`,  
`test_wasb_runner`, `test_reingest_safety`, `test_detector_keying`, `test_config_wasb`,  
`test_path_safety`, `test_reliability_hardening` (+ collector `test_licensing_gate`,  
`test_import_guard`, `test_cvat_robustness`).
```

```
8
9      **LEGEND**
10     - `@path` = file (always repo-relative)
11     - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
12     - `->` = dataflow / call / "produces"
13     - `$` = data contract (canonical shape)
14     - `?` = gotcha / footgun
15     - `?'` = registry / plugin extension point
16     - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
17
18     ---
19
20     ## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
21
22     Top-level tree and the **placement rule** for each kind of file. When unsure, prefer an
existing
23     **registry / extension point** (`?`) over a new top-level file.
24
25     | You're adding? | Put it in? | Notes |
```

```

26 |---|---|---|
27 | Backend / runtime code | `backend/<subsystem>/` |
`pipeline/{segmenters,detectors,validators}`, `providers/`, `sports/`, `annotations/`,
`infrastructure/`, `reporting/`, `stitcher/`, `utils/`, `config/`, `features/`, `api/`. Most
additions are register a class in the relevant registry, not a new module tree. |
28 | Eval LIBRARY (importable, pure ? metrics, calibration, a scorer, a feature) |
`backend/eval/` | Pure core, no side-effects at import; CLI/I-O lives in a separate driver.
These are imported widely (e.g. `calibrate_wasb` by 8 sites) ? they are library, not
scripts; do NOT move them out. |
29 | Standalone run-driver / one-off entry CLI (has __main__, NOT imported by app
code) | scripts/ | e.g. `scripts/run_phase3_eval.py`, `scripts/run_process_and_stitch.py`.
Run as `python scripts/<name>.py`. If it needs load_dotenv() / sys.path before importing
backend, add "scripts/<name>.py" = ["E402"] to ruff.toml. |
30 | Ops / maintenance tool (not part of the served app ? cache GC, batch admin) |
`tools/` | Run as python -m tools.<name> (it's an importable package). e.g.
`tools/cleanup_caches.py`. Keep the destructive decision path pure + unit-tested. |
31 | Training code (model fitting) | `training/` | Behind the CI-enforced import wall
(ADR-008) ? backend.main must not import it. Own requirements-train.txt. |
32 | A test | tests/test_*.py | pytest auto-discovers; the full-suite lane has a 70%
coverage floor. Pure/offline + mocked (no GPU/WSL/network). |
33 | A doc (design, plan, living log) | docs/ | Index it in docs/README.md. Only
terminal-state (DONE/REJECTED) work moves to docs/archives/. |
34 | A throwaway repro / debug script | scratch/ | Gitignored + excluded from
ruff/pytest. Never imported by app code. |
35 | A model artifact (weights + manifest) | artifacts/<name>/<ver>/ | ADR-008:
manifest committed (provenance), weights gitignored. |
36 | Eval baseline / leaderboard JSON | eval_baselines/ | Tracked (survives a clean
checkout); the no-regression gate reads it. |
37 | Committed regression fixture (video-free golden) |
eval_baselines/fixtures/<slug>/ | Tracked, LF-pinned (.gitattributes). Synthetic Tier-0
(or owner-cleared, de-attributed real). Drives the golden_fixtures characterization gate with
no video/GPU/DB. See docs/GOLDEN_REGRESSION_FIXTURES.md; mint via python -m
backend.eval.golden_fixtures --freeze-synthetic / --freeze-real (refusal-gated). |
38 | Pipeline run outputs (reels, trajectories, DBs, frames) | output/ | Gitignored.
Per-video isolation is planned in PER_VIDEO_OUTPUT_LAYOUT.md. |
39
40 Canonical files that stay at the repo ROOT (referenced by CI / packaging / docs ? do
not move):

```

7. user (2026-06-19T17:18:21.238Z)

```

1 # Annotation & Evaluation
2
3 How to measure how well the pipeline detects rallies, using hand-annotated
4 ground-truth timestamps. The harness scores predictions already stored in the
5 SQLite DB ? it makes no API calls and is fully deterministic, so you can run it
6 as often as you like for free.
7
8 ---
9
10 ## 1. What you annotate (and what it does / doesn't do)
11
12 You annotate, per video, the [start, end] time range of each rally. That's
13 all the evaluation harness needs.
14
15 > Important: these rally timestamps measure and tune rally segmentation
16 > ? i.e. where the pipeline says rallies begin and end. They do not train
17 > the shuttle detector (TrackNet/WASB), which learns from per-frame shuttle
18 > (x, y) coordinate labels, a different and much heavier annotation. Use rally
19 > timestamps as your golden evaluation set; only invest in frame-level
20 > coordinate labels if evaluation shows detection (not segmentation) is the
21 > bottleneck. See docs/TRACKNET_WSL_SETUP.md.
22
23 ## 2. Annotation file format (CSV)
24

```

```

25     One CSV per video, one rally per row:
26
27     ```csv
28     start,end
29     00:01:12,00:01:39
30     102.5,131.0
31     ```
32
33     - The `start,end` header is optional (auto-detected).
34     - Timestamps may be decimal seconds (`102.5`) or colon-separated
35       `MM:SS` / `HH:MM:SS` (`00:01:12`). Forms can be mixed.
36     - Blank lines and `#` comments are ignored.
37     - Malformed rows (bad timestamp, `start >= end`) fail loudly rather than being
38       silently skipped ? so labeling mistakes surface immediately.
39
40     Generate an empty template to fill in:

```

8. user (2026-06-19T17:18:21.903Z)

```

1     # How rally detection works (the 2-layer mental model)
2
3     A short conceptual map of how we go from raw video to rally `[start, end]` segments ?
4     and, crucially, which layer our annotation can improve.
5
6     ```
7     video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
8           ?                               ?
9           (WASB CNN,                      per-frame trajectory
10            frozen weights)                ?
11                                           ?
12            velocity signal (how fast the shuttle moves)
13                                           ?
14            [SEGMENTATION] ??? the 3 knobs we tune
15                                           ?
16                                           ?
17            rally windows [start, end] ??? IoU vs golden
labels
18     ```
19
20     ## Two layers ? keep them separate in your head
21
22     **1. DETECTION ? "where is the shuttle, this frame?"**
23     WASB is a CNN that, for each frame, outputs a heatmap ? the shuttle's `(x, y)` + a
24     visibility/confidence. Its weights were learned (by gradient descent, on shuttle-
coordinate
25     labels) and we use them frozen. We chose a *shuttle* detector on purpose: "shuttle
in
26     flight" is the signal for "a rally is happening," and it's camera-agnostic ? unlike
27     player-motion, which fooled the old heuristic on broadcast cameras that pan with the
players.
28     (Your videos are static tripod, so player-motion would also work ? but shuttle-motion is
the
29     robust general bet.)
30
31     **2. SEGMENTATION ? "when is a rally?"**
32     From the per-frame `(x, y)` we compute the shuttle's velocity (displacement between
frames).
33     Then three knobs turn that into rally intervals:
34     - **`velocity_thresh`** ? a frame is "active" only if the shuttle is moving faster than
this
35     (in flight, not lying on the floor / being picked up).
36     - **`merge_gap`** ? active frames within this many seconds get joined into one rally
(bridges
37     brief gaps: a shuttle momentarily occluded, or the beat between two hits).
38     - **`min_window_duration`** ? throw away windows shorter than this (blips / false

```

flickers).

39

40 Output: a list of `[start, end]` windows = the rallies.

9. assistant (2026-06-19T17:18:23.782Z)

These three are clearly reference/navigation docs. Now let me read the remaining four headers.

10. user (2026-06-19T17:18:25.224Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
25     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
```

```

27
28     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
29
30     ## Where we are (one paragraph)
31     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
32     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
33     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
34     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
35     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
36     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
37     battery is done** (see table) ? the moat is quantified.
38
39     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
40     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
41     |---|---|---|
42     | Heuristic (velocity windowing) | 0.657 | 0.157 |
43     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
44     | Two-stage WASB+Gemini refine | 0.83 | ? |
45     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
46
47     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
48     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
49     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
50     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips

```

11. user (2026-06-19T17:18:25.675Z)

```

1     # Roadmap ? from annotated data to a self-improving, offline rally segmenter
2
3     This document is the **narrative thread**: how the indexer evolves from an
4     AI-API-dependent prototype into a self-contained tool that learns rally detection
5     from its own commercially-licensed, annotated video corpus. It explains *how we
6     got here*, *why each step matters*, and *how it shapes the final tool*. Decisions
7     are recorded as ADRs in [DECISIONS.md](archives/decisions/DECISIONS.md); the scoring
mechanics live
8     in [EVALUATION.md](EVALUATION.md).
9
10    > ? **Phase 4 update:** the authoritative owned-model path is now
11    > [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) ? a learned
**per-frame TAL head**
12    > (ActionFormer/ASFormer over WASB trajectory features), not the candidate-window
classifier sketched below.
13    > **Audio is parked, not "gated on golden labels":** the E1 probe and a follow-up
foreground-audio test both came back
14    > weak (~0.6 AUC), so it is de-prioritized even though labels now exist (see the plan's
Gen-1 note).
15
16    ## The thesis
17
18    The indexer detects badminton rallies in long recordings. Its strongest mode
19    (`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
20    player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
21    tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That

```

```

22     works, but it costs money, needs the network, and never improves on its own.
23
24     > *Naming note:* the `yolo_hybrid` registry key is retained for back-compat, but
25     > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
26     > ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local
27     > player-motion gating stage, not the Ultralytics library.
28
29     A sibling project, **sports-data-collector**, curates sports videos with
30     per-rally `[start, end)` annotations and a commercial-license flag. The thesis of
31     this roadmap: **use that annotated, commercially-usable data to (1) measure the
32     pipeline objectively and (2) train an offline replacement for the paid Gemini
33     stage** ? turning the tool into one that improves as more data is curated, runs
34     air-gapped, and generalizes across sports.
35
36     Why it matters: it removes the per-run API cost and network dependency, gives us
37     a defensible quality metric instead of eyeballing, and creates a flywheel ?
38     more curated matches ? a better offline segmenter ? better highlights.
39
40     ## The four phases
41
42     ### Phase 1 ? Bridge the repos (DONE)
43     A sport-agnostic `curate export` in the collector emits the indexer's `start,end`
44     ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on
45     the `is_commercial` flag. ? *Why:* one clean producer/consumer contract that works

```

12. user (2026-06-19T17:18:26.618Z)

```

1     # Backlog ? deferred items to revisit
2
3     Single tracker for work we consciously deferred, so nothing gets lost between sessions.
4     Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry
5     traces to a real decision/PR/doc (no speculative work).
6
7     > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
8     of these to **GitHub issues** instead (or in addition).
9
10    > Status legend: ? blocks something . ? do when triggered . ? nice-to-have/cleanup.
11
12    _Last updated: 2026-06-10._
13
14    ---
15
16    ## Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
17    `NEXT_STEPS.md`)
18
19    - ? **Rename the repo off "badminton"*** ? owner-emphatic "ASAP" action; the project is
20    multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. **Revisit:**
21    owner action, before broad external sharing.
22
23    - ? **Quantify cost-to-Gen-2 (the largest open risk)** ? benchmark real VRAM/token/$ for
24    the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUs)
25    **before** standing up any owned serving stack. **Revisit:** before Phase-2 commitment.
26
27    - ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized
28    (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4
29    harness.
30
31    ## Engineering hardening deferred from the 2026-06-10 audit sweep
32
33    - ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app`
34    leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
35    (#16). **Revisit:** when the async slice starts.
36
37    - ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
38    `*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
39    **Revisit:** before any volume/batch processing.
40
41    - ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):**
42    `detectors/base.py::wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and
43    tilde expansion coexist; applied to every `~`-capable arg in both runners;
44    `tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach
45    `_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the

```

```

default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations
(repo_dir/conda_env) if config ever becomes per-tenant.
21
22 ---
23
24 ## Monetization / licensing
25 *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
26
27 - ? **Ultralytics Enterprise pricing** ? get a quote **only if** the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently *not
needed* (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering
YOLOv8.
28 - ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. **Revisit:** before commercial launch.
29 - ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the *code* license ? the distributed *weights/dataset* terms.
Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed
weights.
30 - ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. **Revisit:** when choosing the production LLM step.
31
32 ## Detector swap follow-ups
33 *(source: PR #8, ADR-005)*
34
35 - ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
36 - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but **not yet validated on actual footage** for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
37 - ? **`yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
38
39 ## Golden-set subproject
40 *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
41
42 - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
43 - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. **Unblocked.** **Revisit:** anytime.
44 - ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora**
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the
Roora pilot confirms quality, then build the adapter.
45 - ? **GS-4 oracle model** ? `AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
**Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).

```

13. user (2026-06-19T17:18:27.132Z)

```

1 <!--
2 PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4 A "tracked project doc" is how this repo carries substantial design/project work: it can
be

```

```

5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12       or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15       research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27     ---
28
29     ## 0. TL;DR
30     <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32     ## 1. Problem & goal
33     <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35     ## 2. Decisions locked
36     <Table ? capture answered questions so they are not relitigated.>
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | ? | ? | ? |
41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
43     <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45     ## 4. Design / architecture

```

14. assistant (2026-06-19T17:18:31.615Z)

All seven docs are clearly live: three are reference docs (CODE_MAP, EVALUATION, HOW_RALLY_DETECTION_WORKS), one is a template (PROJECT_DOC_TEMPLATE), and three are active tracking docs (NEXT_STEPS dated 2026-06-13, ROADMAP narrative, BACKLOG dated 2026-06-10). None say COMPLETED/REJECTED. Now let me cross-check inbound references for each filename across the repo, and confirm none have an archived duplicate.

15. user (2026-06-19T17:18:32.699Z)

CLAUDE.md:7
README.md:2
run_eval.py:2
docs\DEFERRED_HARDENING_PLAN.md:1
docs\BACKLOG.md:2
docs\COMMERCIALIZATION.md:2
docs\CODE_MAP.md:1
scripts\README.md:1
docs\GOLDEN_SET_VENDORING.md:2
docs\GOLDEN_DATA_SHARING.md:1
docs\GOLDEN_SET_PHASE1_VIDEOS.md:2
docs\GOLDEN_SET_IMPLEMENTATION.md:1
docs\EXPERIMENT_HARNESS.md:2
docs\GOLDEN_REGRESSION_FIXTURES.md:4
docs\DESKTOP_APP_PLAN.md:1
docs\archives\decisions\DECISIONS.md:1
docs\archives\COMMERCIAL_READINESS_REVIEW.md:2
docs\ANALYZERS_AND_RECONCILERS.md:2
docs\MULTI_SIGNAL_FUSION_PLAN.md:4
docs\NEXT_STEPS.md:1
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:5
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:4
docs\archives\README.md:1
docs\PLATFORM_ARCHITECTURE.md:2
docs\PROJECT_DOC_TEMPLATE.md:1
docs\QUALITY_ITERATIONS.md:2
docs\README.md:2
docs\RALLY_DETECTION_GAP_CLOSURE.md:3
docs\RALLY_QUALITY_RESEARCH.md:7
docs\ROADMAP.md:2
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:3
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:1
backend\annotations\automated_provider.py:2
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md:1
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:1
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md:1
docs\archives\checkpoints\2026-06-eval-baseline\README.md:1
docs\archives\checkpoints\2026-06-logging-reporting\README.md:1
docs\archives\quality-iterations\2026-06-initial\README.md:1
backend\pipeline\detectors\base.py:1
backend\eval\harness.py:1
backend\pipeline\segmenters\fusion_hybrid.py:1
backend\eval\splits.py:1
backend\eval__init__.py:1
backend\eval\training.py:1

Found 88 total occurrences across 45 files.

16. user (2026-06-19T17:18:33.263Z)

Found 11 files

docs\RALLY_QUALITY_RESEARCH.md
docs\RALLY_DETECTION_GAP_CLOSURE.md
docs\BACKLOG.md
CLAUDE.md
docs\DESKTOP_APP_PLAN.md
docs\GOLDEN_REGRESSION_FIXTURES.md
docs\GOLDEN_DATA_SHARING.md
backend\pipeline\segmenters\fusion_hybrid.py
docs\PROJECT_DOC_TEMPLATE.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md

17. assistant (2026-06-19T17:18:37.320Z)

All seven are heavily referenced as live sources (CLAUDE.md, README.md, NEXT_STEPS.md, code, tests). None have an archived duplicate (the Glob showed no `CODE\_MAP-`, `EVALUATION-`, `NEXT\_STEPS-`, `ROADMAP-`, `BACKLOG-`, `PROJECT\_DOC\_TEMPLATE-`, or `HOW\_RALLY\_DETECTION\_WORKS-` files in archives). Let me get per-doc reference counts to populate the structured output accurately.

18. user (2026-06-19T17:18:38.458Z)

Found 13 files
docs\CODE_MAP.md
docs\BACKLOG.md
CLAUDE.md
docs\GOLDEN_REGRESSION_FIXTURES.md
backend\pipeline\detectors\base.py
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\ROADMAP.md
scripts\README.md
docs\PLATFORM_ARCHITECTURE.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md

19. user (2026-06-19T17:18:38.919Z)

Found 14 files
docs\README.md
docs\RALLY_QUALITY_RESEARCH.md
docs\QUALITY_ITERATIONS.md
docs\GOLDEN_REGRESSION_FIXTURES.md
docs\ROADMAP.md
docs\EXPERIMENT_HARNESS.md
README.md
docs\GOLDEN_SET_VENDORIZING.md
backend\eval\harness.py
docs\archives\decisions\DECISIONS.md
run_eval.py
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\GOLDEN_SET_IMPLEMENTATION.md
backend\eval__init__.py

20. user (2026-06-19T17:18:39.756Z)

Found 9 files
docs\RALLY_QUALITY_RESEARCH.md
docs\RALLY_DETECTION_GAP_CLOSURE.md
docs\QUALITY_ITERATIONS.md
docs\NEXT_STEPS.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
CLAUDE.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\EXPERIMENT_HARNESS.md

21. user (2026-06-19T17:18:40.230Z)

Found 6 files
CLAUDE.md
docs\PLATFORM_ARCHITECTURE.md
docs\COMMERCIALIZATION.md
README.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
backend\eval\training.py

22. user (2026-06-19T17:18:40.693Z)

Found 16 files

CLAUDE.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
backend\eval\splits.py
docs\COMMERCIALIZATION.md
docs\DEFERRED_HARDENING_PLAN.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
backend\annotations\automated_provider.py

23. user (2026-06-19T17:18:41.161Z)

Found 2 files

docs\README.md
CLAUDE.md

24. assistant (2026-06-19T17:18:44.700Z)

All evidence is clear. Every doc in my batch is live, heavily cross-referenced, and has no archived duplicate. All seven are KEEP_LIVE ? three reference docs, one template, and three actively-tracked roadmap/backlog/next-steps docs. None carry a COMPLETED/REJECTED/SUPERSEDED status header.

25. user (2026-06-19T17:19:19.168Z)

Structured output provided successfully

26. assistant (2026-06-19T17:19:27.663Z)

All seven docs in my batch are KEEP_LIVE. Audit complete.

Summary of findings:

- **CODE_MAP.md** ? KEEP_LIVE (reference/navigation; CLAUDE.md mandates as cold-start reading; 13 referrers)
- **EVALUATION.md** ? KEEP_LIVE (reference how-to for the eval harness; linked from live code `backend/eval/harness.py`, `run\_eval.py`; 14 referrers)
- **HOW_RALLY_DETECTION_WORKS.md** ? KEEP_LIVE (policy-named reference doc; head of the live quality track in CLAUDE.md; 9 referrers)
- **NEXT_STEPS.md** ? KEEP_LIVE (self-labeled "(live)", updated 2026-06-13, active GPU-box pickup section; the canonical next-steps tracker; 11 referrers)
- **ROADMAP.md** ? KEEP_LIVE (living narrative roadmap with active Phase-4 pointer; 6 referrers)
- **BACKLOG.md** ? KEEP_LIVE (living deferred-work tracker, mostly open items; 16 referrers)
- **PROJECT_DOC_TEMPLATE.md** ? KEEP_LIVE (workflow-required template; 2 referrers)

No ARCHIVE_NOW, no ASK_OWNER, no DEDUPE_STUB. None of these docs carry a COMPLETED/REJECTED/SUPERSEDED status header, none have an already-archived copy of their stem in `docs/archives/`, and all are cited as live/authoritative sources in CLAUDE.md, README.md, and/or live code ? so none are in a terminal state and archiving any would break inbound links.